Bipolaris-RNAi Designer (BRD) v3 — Validation and Execution Report

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Execution Mode: Full 8-Stage Computational Research Release (`11_post_execution_audit_v3`)

Mandatory Scientific Disclosure: All scores, Pareto tiers, and candidate shortlists in this report are **computational prioritizations** across genomic/transcriptomic datasets. Nothing in this report claims demonstrated fungal dsRNA uptake, in vivo gene knockdown, crop protection against *Bipolaris sorokiniana*, or environmental biosafety.

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1. Research objective and computational scope

Bipolaris-RNAi Designer v3 upgrades the prior v1 (`10_final_reports/`) and v2 (`11_post_execution_audit/`) computational RNAi candidate prioritisation pipeline targeting **4* Bipolaris sorokiniana *(syn.* Cochliobolus sativus*) pathogenicity/biology genes* ***(`CsFUS3` , `CsSTE11` , `CsSp1` , `CsPKS1`) across*** *19 public* B. sorokiniana *genome assemblies**.

Key methodological upgrades in v3:

- Reconciles all v1/v2 numerical and filename discrepancies (`reconciliation_log_v3.tsv`) and locks **1 authoritative dataset** (`authoritative_data/`).
- Fixes the M04 stale `.dmnd` cache bug in `brd/integrations/diamond.py` , recovering real core-proteome orthology and distinguishing annotation gaps from gene loss.
- Separates unpolished long-read homopolymer frameshift artifacts (`WAI2406` , `WAI2411`) from genuine biological polymorphisms across all 50 dsRNA candidates.
- Expands non-target screening from 7 transcriptomes to **17 non-target organisms** (4 crops, 9 beneficial fungi with whole genomes, 4 soil/rhizosphere bacteria) plus self-*B. sorokiniana* paralogs/genome on **both strands**.
- Clusters the 50 overlapping M07 windows into **12 non-redundant independent target regions**.
- Enforces strict **3-dimensional separation** (`biological_target_relevance_score` , `dsrna_targetability_score` , `sequence_selectivity_score`), Pareto optimality, weight sensitivity, isolate bootstrapping, and primary-source literature benchmarking.

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2. Data sources and accession provenance

Authoritative Target Loci (`auth_targets.tsv`)

target	locus_tag	refseq_protein_id	refseq_mrna_id	uniprot_accession	cds_length_nt	verification_status
CsPKS1	COCSADRAFT_157080	XP_007696213.1	XM_007698023.1	M2RQ22	6462	Confirmed (Primer + Full RefSeq/GenBank Verified)
CsFUS3	COCSADRAFT_24574	XP_007697446.1	XM_007699256.1	M2TCC8	1059	Confirmed (Primer + Full RefSeq/GenBank Verified)
CsSTE11	COCSADRAFT_105370	XP_007694574.1	XM_007696384.1	M2TIY7	2880	Confirmed (Primer + Full RefSeq/GenBank Verified)
CsSp1	COCSADRAFT_164106	XP_007704210.1	XM_007706020.1	M2RXZ0	270	Confirmed via 5/5 Primers + Reconstructed 270-nt RefSeq CDS (No standalone nuccore accession)

Reconciliation of Prior v1/v2 Discrepancies (`reconciliation_log_v3.tsv`)

discrepancy_id	entity	earlier_value	later_v2_value
REC_01_STE11_CAND012_COMPOSITE	BRD_CsSTE11_CAND012	v1=0.7832; audit(2).zip=0.8744	0.917
REC_02_STE11_CAND013_COMPOSITE	BRD_CsSTE11_CAND013	v1=0.7321; audit(2).zip=0.8633	0.9059
REC_03_CSFUS3_BIO_SCORE	CsFUS3 biological_target_score	audit(2).zip=0.9034	0.9915
REC_04_CSSTE11_BIO_SCORE	CsSTE11 biological_target_score	audit(2).zip=0.8837	0.9866
REC_05_CSSP1_BIO_SCORE	CsSp1 biological_target_score	audit(2).zip=0.8904	0.6993
REC_06_CSSP1_EXPRESSION_SPIKE	CsSp1 GSE130517 RPKM	v1/infection_expression.tsv=1438.44 (functional_evidence_summary=1438.44)	valid 3B/4B/5B max=139.61 (sum raw reads=75); 3A_I RPKM=1438.44 from raw_reads=1
REC_07_TEST_PATHS_AND_FILENAME	test_audit_v2.py & archives	audit(3).zip test: ['OUT_DIR = Path(r"C:\\Users\\ASUS BANGLADESH\\Desktop\\New folder (2)\\Output")', 'near = pd.read_csv(AUDIT_DIR / "off_target_near_matches_v2.tsv", sep="\\t")']	archive near= ['11_post_execution_audit/off_target_near_guide_matches_v2.tsv']
REC_08_M04_CORE_ACCESSORY_CACHE_BUG	04_comparative_genomics/core_accessory & diamond.py	v1 core orthogroups=0/20288	Bsor_Sacc._Shoemaker absent from all_proteomes_db.dmnd subjects (0/249,075 subject hits)

Pipeline Inputs & Software Versions (`software_and_reference_versions_v3.tsv`)

- **Isolate panel:** `19` *B. sorokiniana* assemblies (`GCF_000338995.1` RefSeq `ND90Pr` + 18 GenBank assemblies; `auth_isolates.tsv`).
- **Functional databases:** PHI-base v4.19, DEG essential genes, InterPro (`05_functional_evidence/`), NCBI GEO `GSE130517` barley-root RNA-seq.

component	version_or_status	notes
Python	3.13.14	Windows-11-10.0.26300-SP0
biopython	1.86	Sequence parsing & semi-global alignments
pandas / numpy / scipy	2.3.3 / 2.4.0 / 1.16.3	Statistical tables, Spearman, bootstrap
ViennaRNA	2.8.0	Duplexfold MFE & whole-candidate RNA folding
matplotlib / networkx	3.10.8 / 3.6.1	Publication figures & RBH/overlap clustering
BLAST+ (blastn/tblastn/makeblastdb)	blastn: 2.17.0+	Off-target blastn/blastn-short & core marker tblastn
DIAMOND	diamond version 2.2.8	Repaired all-vs-all RBH orthology clustering
NCBI datasets CLI	datasets version: 18.41.0	Expanded reference genome & transcriptome retrieval
MAFFT	Active code page: 65001	M04/M07 baseline multiple sequence alignments
dsRNAmax	2026/11/11 12:59:18 dsRNAmax - dsRNA maximizer (Version: 1.1.14)	Included in M07 baseline provenance
QUAST / BUSCO / OrthoFinder	NOT_RUN_DEPENDENCY_BLOCKED	Linux/WSL2/Conda dependencies absent on Windows host; zero outputs fabricated
RNAhybrid	NOT_RUN_DEPENDENCY_BLOCKED	Linux-only Bioconda binary; ViennaRNA 2.8.0 used as native comparator

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3. Genome QC and assembly limitations

All 19 assemblies span `32.43–38.24 Mb` and `49.24–50.25%` GC (`assembly_metrics_19_isolates_v3.tsv`), confirming conspecificity and ruling out large-scale genome contamination. However, two structural assembly limitations materially impact downstream inference:

1. **Scaffold fragmentation & N-gaps:** `ND90Pr` (`154` scaffolds, `3.48%` N-bases = `~1.19 Mb`) and `Sacc._Shoemaker` (`GCA_013416765.1` , `3.48%` N-bases) contain extensive scaffolding gaps, partially truncating flanking or edge alignments.
2. **Unpolished long-read homopolymer indel frameshifts (`WAI2406` , `WAI2411`):** Testing 50 conserved single-copy core fungal markers via `tblastn` establishes an exon-intron boundary HSP stop baseline of `52–54` stops (`26–27` markers) across `ND90Pr` and all `16` remaining high-quality assemblies (`HQ-17`). By contrast, `WAI2406` (`190` stops, `+137` excess over `ND90Pr`) and `WAI2411` (`235` stops, `+182` excess) exhibit systematic frameshift disruptions across conserved housekeeping loci despite 50/50 marker detection (`isolates_artifact_vs_quality_v3.tsv`). Therefore, indels inside `WAI2406` / `WAI2411` cannot be treated as true biological gene loss.
3. **External Linux QC tool blockage:** `QUAST` and `BUSCO` did **not** run natively on Windows (no WSL2 Linux distro present) and are never substituted or faked (`Section 16`).

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4. Comparative genomics results

1. **Stale `.dmnd` Cache Repair (`brd/integrations/diamond.py`):** M04 previously reported `0/20,288` core orthogroups shared with `Bsor_Sacc._Shoemaker` because `build_diamond_db` skipped rebuilding an outdated `all_proteomes_db.dmnd` . After repairing the DB rebuild and running clean `DIAMOND v2.2.8` all-vs-all RBH clustering across `Bsor_ND90Pr` (`12,210` proteins), `Bsor_Sacc._Shoemaker` (`10,755` proteins), and `Botrytis cinerea ASM83294v1` , **`7,245` core orthogroups** are cleanly recovered across both *B. sorokiniana* isolates.
2. **Annotation Gap vs Gene Loss (`annotation_comparability_audit_v3.tsv`):**

metric	count	interpretation
ND90Pr total annotated proteins	12210	NCBI RefSeq GCF_000338995.1 (JGI Cocsa1 model)
Sacc._Shoemaker total annotated proteins	10755	NCBI GenBank GCA_013416765.1 (1,455 fewer protein models than ND90Pr)
v1/v2 stale-cache core orthogroups (ND90Pr + Sacc.)	0	BUG: old all_proteomes_db.dmnd omitted Sacc._Shoemaker subjects
v3 repaired Core orthogroups (ND90Pr & Sacc._Shoemaker present)	7245	Genuine conserved core proteome shared across both annotated B. sorokiniana genomes
Orthogroups annotated only in ND90Pr (Sacc._Shoemaker proteome = 0)	2162	Of 2385 such ND90Pr proteins, 2263 (94.9%) match Sacc._Shoemaker raw genome (>=70% aa tblastn) -> primarily unannotated gene models/gaps, not genuine gene loss!

Of `2,385` `ND90Pr` -only proteins lacking a `Sacc._Shoemaker` proteome annotation, **`tblastn` recovers `2,263` (`94.9%`) directly in the raw `Sacc._Shoemaker` genome assembly (`>=70%` aa identity)**, proving the discrepancy reflects annotation incompleteness rather than biological deletion.

1. **OrthoFinder status:** Linux `OrthoFinder` (`mcl` / `fastme`) is blocked on Windows (`NOT_RUN_DEPENDENCY_BLOCKED`); v3 explicitly labels its orthology table as `DIAMOND_v2.2.8_AllVsAll_RBH` .

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5. Conservation across isolates

Evaluating conservation using **minimum identity and exact 21-mer preservation across `HQ-17` isolates** separately from `WAI2406` / `WAI2411` prevents single-isolate indels or high means from masking divergence (`candidate_conservation_multimetric_v3.tsv`):

- **Total candidates assessed:** `50` across 4 targets; `50/50` achieve complete detection across the 19 assemblies (`2/19` isolates flagged `Unresolved_Assembly_Polishing_Artifact` for frameshift indels).

- **Mean vs Minimum disparity:** Several candidates maintain high mean identity (`>98%`) while dropping in `min_21mer_preservation_hq17_pct` or `min_21mer_preservation_all19_pct` due to single-nucleotide polymorphisms or `WAI2406/WAI2411` indels.

Conservation Threshold Sensitivity (`conservation_threshold_sensitivity_v3.tsv`)

scenario_id	evaluated_metric	threshold_pct	passing_candidates_n	CsSTE11_pass	CsPKS1_pass	CsSp1_pass	CsFUS3_pass
Scenario_A1_Mean21mer_ge85pct	mean_21mer_preservation_all19_pct	85	48	12	13	10	13
Scenario_A2_Mean21mer_ge90pct	mean_21mer_preservation_all19_pct	90	46	10	13	10	13
Scenario_A3_Mean21mer_ge95pct	mean_21mer_preservation_all19_pct	95	44	8	13	10	13
Scenario_B1_MinNtID_All19_ge85pct	min_nt_identity_all19_pct	85	50	14	13	10	13
Scenario_B2_MinNtID_All19_ge90pct	min_nt_identity_all19_pct	90	50	14	13	10	13
Scenario_B3_MinNtID_All19_ge95pct	min_nt_identity_all19_pct	95	50	14	13	10	13
Scenario_C1_MinNtID_HQ17_ge90pct	min_nt_identity_hq17_pct	90	50	14	13	10	13
Scenario_C2_MinNtID_HQ17_ge95pct	min_nt_identity_hq17_pct	95	50	14	13	10	13
Scenario_C3_Min21mer_HQ17_ge95pct	min_21mer_preservation_hq17_pct	95	29	6	12	0	11
Scenario_D1_90pctOfIsolates_ge95pctNtID	isolates_ge95pct_nt_id_pct	90	50	14	13	10	13

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6. Off-target screening coverage and findings

The v3 screen expands from M08's 7 transcriptomes to **18 reference entries** (`17` non-target organisms + `1` self *B. sorokiniana* `ND90Pr` control with target loci masked; `expanded_reference_manifest_v3.tsv` , `reference_panel_coverage_and_gaps_v3.tsv`):

- **Crops (`4`):** Wheat (`T. aestivum`), Barley (`H. vulgare`), Rice (`O. sativa` `GCF_034140825.1` tx+genome), Maize (`Z. mays` `GCF_902167145.1` tx).
- **Beneficial Fungi (`9`):** *Trichoderma atroviride*, *T. virens*, *T. reesei*, *T. harzianum* (`GCF_003025095.1`), *Beauveria bassiana*, *Metarhizium anisopliae* (`GCF_039654215.1`), *M. robertsii* (`GCA_000591435.1`), *Rhizophagus irregularis*, *R. clarus* (`GCA_003203555.1`), all including **whole-genome dual-strand 21-mer scans**.
- **Rhizosphere/Soil Bacteria (`4`):** *Bacillus subtilis* (`GCF_000009045.1`), *B. velezensis* (`GCF_000508125.2`), *Pseudomonas fluorescens* (`GCF_900215245.1`), *P. putida* (`GCF_000412675.1`).
- **Passing zero-exact-21mer hard filter:** `32/50` (`18/50` excluded by `>=1` exact dual-strand 21-mer match across the expanded panel!).

Expanded Selectivity Classification Summary (`candidate_selectivity_expanded_v3.tsv`)

selectivity_classification_v3	candidates_n
Elevated_20of21_Near_Match_Burden	29
Exact_21mer_NonTarget_Or_Paralog_Match	18
Low_20of21_Near_Match_Signal	3

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7. Candidate redundancy and independent target regions

Many of the original 50 M07 candidates were overlapping sliding/multi-tool windows (`SiFi21` , `dsRNAbuilder` , `dsRNAmax`) on identical CDS loci. Clustering windows sharing `>=25%` CDS overlap or `>=25%` 21-mers reduces **50 candidates into `12` non-redundant independent target regions** (`independent_target_regions_v3.tsv`):

region_id	target	cds_span_start_nt	cds_span_end_nt	region_span_nt	n_redundant_candidates	n_passing_clean_hq85	design_tools_present
CsFUS3_REG01	CsFUS3	1	300	300	5	0	SIREN;dsRNAmax
CsFUS3_REG02	CsFUS3	326	710	385	7	0	BRD_IsolateAware;LUCID_sifi21_plus
CsFUS3_REG03	CsFUS3	676	875	200	1	0	LUCID_sifi21_plus
CsPKS1_REG01	CsPKS1	1	300	300	5	5	SIREN;dsRNAmax
CsPKS1_REG02	CsPKS1	4526	4735	210	1	1	BRD_IsolateAware
CsPKS1_REG03	CsPKS1	5651	5830	180	1	1	BRD_IsolateAware
CsPKS1_REG04	CsPKS1	5951	6210	260	6	6	BRD_IsolateAware;LUCID_sifi21_plus
CsSTE11_REG01	CsSTE11	1	300	300	8	6	BRD_IsolateAware;LUCID_sifi21_plus;SIREN;dsRNAmax
CsSTE11_REG02	CsSTE11	1076	1285	210	2	2	BRD_IsolateAware
CsSTE11_REG03	CsSTE11	1401	1610	210	1	1	BRD_IsolateAware
CsSTE11_REG04	CsSTE11	2226	2525	300	3	0	LUCID_sifi21_plus
CsSp1_REG01	CsSp1	1	255	255	10	10	BRD_IsolateAware;LUCID_sifi21_plus;SIREN

- **Exhaustive 180-nt Full-CDS Sweep (`exhaustive_cds_180nt_sweep_v3.tsv`):** Scanning `25-nt` steps across all 4 ORFs shows only `5` clean 180-nt windows on `CsFUS3` (`1,059 nt`) escape exact 21-mer homology with beneficial fungi, proving that MAPK kinase conservation spans virtually the entire `CsFUS3` gene (rather than being an M07 sampling quirk). Outside M07 windows, `112` clean high-conservation exploratory windows (`HQ-17 >= 95%` , `0` exact 21-mers across non-targets) were identified on `CsSTE11` / `CsPKS1` .

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8. Biological target evidence

Biological target relevance (`biological_target_relevance_score`) is scored strictly at the gene level with context-aware PHI-base evaluation and valid `3B/4B/5B` RNA-seq reporting excluded from composite equations (`biological_target_relevance_v3.tsv`):

target	biological_target_relevance_score	subscore_direct_ko_pathogenicity	subscore_phi_context_aware	subscore_sequence_traceability	gse130517_valid_3b4b5b_mean_rpkm_separate
CsFUS3	0.9921	1	1	1	62.53
CsSTE11	0.9921	1	1	1	22.12
CsPKS1	0.5796	0.25	0.25	1	13.19
CsSp1	0.55	1	0	0.5	79.3

- Crucial biological context: **`CsPKS1`** ranks **`4th`** (`0.5796`) in target relevance because its direct *B. sorokiniana* PHI-base entry `PHI:2260` (`Pks1`) explicitly documents **`unaffected pathogenicity`** (affecting DHN spore melanization, not host virulence!).

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9. Candidate targetability

Candidate targetability (`dsrna_targetability_score`) isolates sequence-intrinsic properties (`min_nt_identity_hq17_pct` , `min_21mer_preservation_hq17_pct` , `guides_pct_preserved_hq17` , `efficient_sifi21_fraction` , `guide_acc_mean` , and sequence entropy/complexity) with zero biological-target leakage and zero tool-agreement weighting.

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10. Selectivity assessment

All off-target counts (`exact_21_crops_total` , `exact_21_beneficial_fungi_tx` , `exact_21_beneficial_fungi_gen` , bacteria, self-paralog/genome, `20/21` near matches, `>=30 nt` alignments) feed `sequence_selectivity_score` (`heuristic_sequence_selectivity_index`). Zero exact 21-mer hits across covered panels does **not** prove environmental safety because wheat, barley, and maize whole non-coding genomes (~21 GB combined) and unsequenced soil/phyllosphere biota remain outside local coverage (`reference_panel_coverage_and_gaps_v3.tsv`). Conversely, insect 21-nt feeding comparators (e.g., *Diabrotica* Bachman et al. 2013) are explicitly distinguished as cross-kingdom comparators rather than fungal silencing thresholds.

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11. Pareto and sensitivity analysis

- **3D Pareto Optimal Tiers:** Evaluated both across all 50 candidates (`pareto_front_3d_all50`) and across the `12` **independent target region representatives** (`pareto_front_3d_regions`). Among passing independent regions (`n=8`), Pareto Tier-1 captures candidates trading off higher `CsSTE11` / `CsSp1` biological relevance against higher `CsPKS1` / `CsSTE11` sequence selectivity.
- **Weight Sensitivity (`stage6_weight_sensitivity_spearman_v3.tsv`):**

scenario_a	scenario_b	spearman_rho	p_value
S1_Balanced_30_35_35	S2_SelectivityFirst_20_30_50	1	0
S1_Balanced_30_35_35	S3_BioPathogenicityFirst_50_25_25	1	0
S1_Balanced_30_35_35	S4_TargetabilityFirst_20_50_30	0.9286	0.000863
S1_Balanced_30_35_35	S5_EqualWeights_33_33_33	1	0
S2_SelectivityFirst_20_30_50	S3_BioPathogenicityFirst_50_25_25	1	0
S2_SelectivityFirst_20_30_50	S4_TargetabilityFirst_20_50_30	0.9286	0.000863
S2_SelectivityFirst_20_30_50	S5_EqualWeights_33_33_33	1	0
S3_BioPathogenicityFirst_50_25_25	S4_TargetabilityFirst_20_50_30	0.9286	0.000863
S3_BioPathogenicityFirst_50_25_25	S5_EqualWeights_33_33_33	1	0
S4_TargetabilityFirst_20_50_30	S5_EqualWeights_33_33_33	0.9286	0.000863

- **1,000-Iteration HQ-17 Isolate Bootstrap (`stage6_bootstrap_stability_v3.tsv`):**

candidate_id	target	region_id	observed_clean_region_rank	bootstrap_mean_rank	bootstrap_median_rank	bootstrap_rank_sd	bootstrap_rank_ci95_low	bootstrap_
BRD_CsSTE11_CAND013	CsSTE11	CsSTE11_REG01	1	1	1	0	1	1
BRD_CsSTE11_CAND014	CsSTE11	CsSTE11_REG03	2	2	2	0	2	2
BRD_CsSTE11_CAND010	CsSTE11	CsSTE11_REG02	3	3	3	0	3	3
BRD_CsPKS1_CAND012	CsPKS1	CsPKS1_REG02	4	4	4	0	4	4
BRD_CsPKS1_CAND010	CsPKS1	CsPKS1_REG04	5	7	7	0	7	7
BRD_CsPKS1_CAND013	CsPKS1	CsPKS1_REG03	6	5	5	0	5	5
BRD_CsPKS1_CAND006	CsPKS1	CsPKS1_REG01	7	6	6	0	6	6
BRD_CsSp1_CAND008	CsSp1	CsSp1_REG01	8	8	8	0	8	8

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12. Literature benchmarking

benchmark_category	study_records
3_Gene_Level_Only	14
5_Ambiguous_or_Conflicting_Outcome	5
4_Computational_Only	4
2_Reconstructed_Sequence	2

Across `25` curated primary records spanning *Fusarium graminearum*, *Botrytis cinerea*, *Sclerotinia sclerotiorum*, *Phakopsora pachyrhizi*, *Magnaporthe oryzae*, *Alternaria alternata*, *Aspergillus flavus*, *Rhizoctonia solani*, and *Bipolaris sorokiniana* (`literature_benchmark_dataset_v3.tsv`), primary full-text XML analysis shows:

1. **Delivery & uptake dependency (`Qiao et al. 2021, PMC8428832` ; `Wytinck et al. 2020, PMC7391711`):** High dsRNA uptake species (*B. cinerea*, *S. sclerotiorum*, *A. niger*, *R. solani*, *V. dahliae*) contrast sharply with zero/weak uptake species (*Colletotrichum gloeosporioides*, *Trichoderma virens*, *P. infestans*). *B. sorokiniana* possesses intracellular RNAi machinery (`Leng et al. 2011`), yet **exogenous dsRNA uptake has not been experimentally demonstrated in* B. sorokiniana***.
2. **Construct length vs SiFi/tool length (`Werner et al. 2020, PMC7202221`):** Manual long constructs (`658–912 nt`) outperformed short tool-designed constructs (`173–193 nt`) on *F. graminearum* `AGO1/2` & `DCL1/2` , including failing (`No silencing`) in two short tool constructs (`AGO1-DCL2` , `AGO1-AGO2`).
3. **Dataset scope limit:** Because primary datasets span heterogeneous delivery routes (SIGS vs HIGS vs liquid culture), variable concentrations, and limited verified full construct sequences (`stage7_benchmark_report_v3.md`), Stage 7 deliberately reports a **structured qualitative + exploratory benchmark** rather than claiming predictive statistical accuracy.

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13. Final prioritized candidate shortlist

The redundancy-reduced computational shortlist selects **1 highest-ranking representative per independent Stage 5 target region** (`8` clean representatives passing the zero-exact-21mer hard filter out of `12` independent regions; full 50-candidate Evidence Cards in `stage6_multidim_ranking/candidate_evidence_cards_v3.md`):

independent_region_rank_v3	candidate_id	target	region_id	cds_coords	candidate_length_nt	biological_target_relevance_score	dsrna_targetability_score	sequenc
1	BRD_CsSTE11_CAND013	CsSTE11	CsSTE11_REG01	1..210	210	0.9921	0.8855	0.955
2	BRD_CsSTE11_CAND014	CsSTE11	CsSTE11_REG03	1401..1610	210	0.9921	0.8859	0.94
3	BRD_CsSTE11_CAND010	CsSTE11	CsSTE11_REG02	1101..1280	180	0.9921	0.8913	0.79
4	BRD_CsPKS1_CAND012	CsPKS1	CsPKS1_REG02	4526..4735	210	0.5796	0.8879	0.895
5	BRD_CsPKS1_CAND010	CsPKS1	CsPKS1_REG04	6001..6180	180	0.5796	0.9009	0.82
6	BRD_CsPKS1_CAND013	CsPKS1	CsPKS1_REG03	5651..5830	180	0.5796	0.8335	0.91
7	BRD_CsPKS1_CAND006	CsPKS1	CsPKS1_REG01	5..254	250	0.5796	0.8772	0.865
8	BRD_CsSp1_CAND008	CsSp1	CsSp1_REG01	1..180	180	0.55	0.7524	0.88

14. Reasons each candidate ranks where it does

All `12` independent target region representatives (`8` passing + `4` excluded for exact 21-mer matches):

- **Rank 1 — `BRD_CsSTE11_CAND013` (`CsSTE11` , `CsSTE11_REG01` , n=8 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.9921, Targ=0.8855 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.955 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=3, long=0). Heuristic index= `0.9418` (3D Region Pareto Tier `1`).
- **Rank 2 — `BRD_CsSTE11_CAND014` (`CsSTE11` , `CsSTE11_REG03` , n=1 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.9921, Targ=0.8859 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.94 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=4, long=0). Heuristic index= `0.9367` (3D Region Pareto Tier `1`).
- **Rank 3 — `BRD_CsSTE11_CAND010` (`CsSTE11` , `CsSTE11_REG02` , n=2 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.9921, Targ=0.8913 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.79 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=14, long=0). Heuristic index= `0.8861` (3D Region Pareto Tier `1`).
- **Rank 4 — `BRD_CsPKS1_CAND012` (`CsPKS1` , `CsPKS1_REG02` , n=1 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.5796, Targ=0.8879 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.895 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=7, long=0). Heuristic index= `0.7979` (3D Region Pareto Tier `1`).
- **Rank 5 — `BRD_CsPKS1_CAND010` (`CsPKS1` , `CsPKS1_REG04` , n=6 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.5796, Targ=0.9009 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.82 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=12, long=0). Heuristic index= `0.7762` (3D Region Pareto Tier `1`).
- **Rank 6 — `BRD_CsPKS1_CAND013` (`CsPKS1` , `CsPKS1_REG03` , n=1 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.5796, Targ=0.8335 (HQ17 min nt=99.44%, min 21-mer=86.88%), Sel=0.91 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=6, long=0). Heuristic index= `0.7841` (3D Region Pareto Tier `2`).
- **Rank 7 — `BRD_CsPKS1_CAND006` (`CsPKS1` , `CsPKS1_REG01` , n=5 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.5796, Targ=0.8772 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.865 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=9, long=0). Heuristic index= `0.7836` (3D Region Pareto Tier `2`).
- **Rank 8 — `BRD_CsSp1_CAND008` (`CsSp1` , `CsSp1_REG01` , n=10 overlapping windows, CLEAN_0_EXACT_21MER):** Bio=0.55, Targ=0.7524 (HQ17 min nt=98.89%, min 21-mer=79.38%), Sel=0.88 (crops 21-mer=0, ben.fungi tx/gen=0/0, 20/21=8, long=0). Heuristic index= `0.7363` (3D Region Pareto Tier `3`).
- **Rank 9 — `BRD_CsFUS3_CAND001` (`CsFUS3` , `CsFUS3_REG02` , n=7 overlapping windows, EXCLUDED_EXACT_21MER):** Bio=0.9921, Targ=0.8958 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.1 (crops 21-mer=0, ben.fungi tx/gen=4/4, 20/21=35, long=14). Heuristic index= `0.6462` (3D Region Pareto Tier `1`).
- **Rank 10 — `BRD_CsFUS3_CAND003` (`CsFUS3` , `CsFUS3_REG03` , n=1 overlapping windows, EXCLUDED_EXACT_21MER):** Bio=0.9921, Targ=0.8836 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.09 (crops 21-mer=1, ben.fungi tx/gen=0/0, 20/21=14, long=4). Heuristic index= `0.6384` (3D Region Pareto Tier `2`).
- **Rank 11 — `BRD_CsFUS3_CAND009` (`CsFUS3` , `CsFUS3_REG01` , n=5 overlapping windows, EXCLUDED_EXACT_21MER):** Bio=0.9921, Targ=0.858 (HQ17 min nt=100.0%, min 21-mer=100.0%), Sel=0.1 (crops 21-mer=0, ben.fungi tx/gen=5/5, 20/21=66, long=11). Heuristic index= `0.6329` (3D Region Pareto Tier `2`).
- **Rank 12 — `BRD_CsSTE11_CAND004` (`CsSTE11` , `CsSTE11_REG04` , n=3 overlapping windows, EXCLUDED_EXACT_21MER):** Bio=0.9921, Targ=0.7789 (HQ17 min nt=99.0%, min 21-mer=78.89%), Sel=0.1 (crops 21-mer=0, ben.fungi tx/gen=2/2, 20/21=24, long=7). Heuristic index= `0.6052` (3D Region Pareto Tier `3`).

15. Candidate-specific weaknesses and uncertainty

- `BRD_CsSTE11_CAND013` (`CsSTE11_REG01`): 3 unique 20/21-mer near-matches
- `BRD_CsSTE11_CAND014` (`CsSTE11_REG03`): 4 unique 20/21-mer near-matches
- `BRD_CsSTE11_CAND010` (`CsSTE11_REG02`): 14 unique 20/21-mer near-matches
- `BRD_CsPKS1_CAND012` (`CsPKS1_REG02`): 7 unique 20/21-mer near-matches; Target biology: B. sorokiniana PHI:2260 Pks1 is 'unaffected pathogenicity' (melanin only)
- `BRD_CsPKS1_CAND010` (`CsPKS1_REG04`): 12 unique 20/21-mer near-matches; Target biology: B. sorokiniana PHI:2260 Pks1 is 'unaffected pathogenicity' (melanin only)
- `BRD_CsPKS1_CAND013` (`CsPKS1_REG03`): 6 unique 20/21-mer near-matches; Target biology: B. sorokiniana PHI:2260 Pks1 is 'unaffected pathogenicity' (melanin only)
- `BRD_CsPKS1_CAND006` (`CsPKS1_REG01`): 9 unique 20/21-mer near-matches; Target biology: B. sorokiniana PHI:2260 Pks1 is 'unaffected pathogenicity' (melanin only)
- `BRD_CsSp1_CAND008` (`CsSp1_REG01`): 8 unique 20/21-mer near-matches; HQ-17 min nt identity dips to 98.89%; Target traceability/length: 270-nt short CDS without standalone nuccore accession

16. Failed, unavailable, or incomplete analyses

Honest inventory (`unavailable_failed_or_incomplete_analyses_v3.tsv`):

tool	status	blocker_or_alternative	score_fabrication_guard
QUAST	NOT_RUN_DEPENDENCY_BLOCKED	Requires Linux/WSL2 + Perl/minimap2/GlimmerHMM; Windows host has no WSL2 distro.	Zero BUSCO/QUAST/OrthoFinder synthetic outputs claimed
BUSCO	NOT_RUN_DEPENDENCY_BLOCKED	Requires Linux/macOS HMMER/MetaEuk/Augustus + fungi_odb10/pleosporales_odb10.	Zero BUSCO/QUAST/OrthoFinder synthetic outputs claimed
OrthoFinder	NOT_RUN_DEPENDENCY_BLOCKED	Requires Linux MCL/FastME/OrthoFinder runtime; mcl & fastme absent natively on Windows.	Zero BUSCO/QUAST/OrthoFinder synthetic outputs claimed
RNAhybrid	NOT_RUN_DEPENDENCY_BLOCKED	Linux/macOS Bioconda binary unavailable natively on Windows; ViennaRNA 2.8.0 RNA.duplexfold used cleanly.	Zero BUSCO/QUAST/OrthoFinder synthetic outputs claimed

Additional incomplete / missing elements documented without substitution:

- `WAI2406` (`GCA_008452705.1`) and `WAI2411` (`GCA_008452715.1`) contain unpolished homopolymer frameshift artifacts (`+137` / `+182` stops above baseline), blocking true coding polymorphism resolution in those 2 assemblies.
- Wheat (`T. aestivum`), barley (`H. vulgare`), and maize (`Z. mays`) whole non-coding nuclear genomes (`>20 GB`) were excluded from local storage; off-target screening covers their complete RefSeq transcriptomes.
- Unsequenced rhizosphere/phyllosphere microorganisms and wet-lab dsRNA uptake in *B. sorokiniana* remain empirically untested.

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17. Reproducibility instructions

All v3 modules resolve relative to repository root (`11_post_execution_audit_v3/src/paths_v3.py` , optional `BRD_OUTPUT_DIR` env override) and contain zero hardcoded machine paths:

```
# 1. Activate environment containing packages listed in 11_post_execution_audit_v3/requirements.txt
# 2. Verify Stages 1-8 end-to-end & execute full test suite:
pytest 11_post_execution_audit_v3/tests/test_brd_v3.py tests/test_brd.py 11_post_execution_audit/tests/test_audit_v2.py -v
# 3. Verify SHA256 integrity manifest:
sha256sum -c 11_post_execution_audit_v3/checksums_v3.txt
```

(See `11_post_execution_audit_v3/README.md` for optional Linux/WSL2 installation of `QUAST` , `BUSCO` , and `OrthoFinder` .)

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18. Explicit distinction between computational predictions and experimentally demonstrated effects

Layer	Status in *Bipolaris sorokiniana*	Source / Basis
Intracellular fungal RNAi pathway	Demonstrated (`CsPKS1` / `GFP` / `ToxA` hairpin RNAi)	`Leng et al. (2011)` (`doi:10.1111/j.1364-3703.2010.00666.x`)
Target gene virulence phenotypes (`CsFUS3` , `CsSTE11` , `CsSp1` , `CsPKS1`)	Demonstrated via gene knockouts / hairpin RNAi	`Leng & Zhong (2015) PMC4443973` ; `Zhang et al. (2022) PMC8743017` ; `Leng et al. (2011) PHI:2260`
Exogenous dsRNA uptake & topical SIGS protection in* B. sorokiniana*	**NOT DEMONSTRATED (untested computational hypothesis)**	Extrapolated cautiously from `Wang et al. (2016)` , `Koch et al. (2016)` , `McLoughlin et al. (2018)` , `Qiao et al. (2021)`
Candidate targetability, conservation & selectivity rankings (`11_post_execution_audit_v3/`)	**100% Computational Prediction Only**	Multi-isolate `tblastn` /semi-global alignments, `blastn` /21-mer scans, ViennaRNA folding, heuristic Pareto indices

Conclusion: `BRD_CsSTE11_CAND013` (Region `CsSTE11_REG01`) and its companion non-redundant representatives (`BRD_CsSTE11_CAND014 [CsSTE11_REG03], BRD_CsSTE11_CAND010 [CsSTE11_REG02], BRD_CsPKS1_CAND012 [CsPKS1_REG02], BRD_CsPKS1_CAND010 [CsPKS1_REG04]`) are **prioritized computational candidates** for future laboratory validation — **not** proven fungicides, silencing reagents, or biosafe treatments.